Optimal Unconventional Policy in a New Keynesian Preferred Habitat Model

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The views expressed do not necessarily reflect the views of the Federal Reserve Bank of Chicago or the Federal Reserve System.

Introduction

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- Since the GFC, central bank balance sheets are an essential tool of monetary policy
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  - How should the two instruments be designed together?
- **This paper:** consensus overlooks welfare costs of both tools with endogenous risk prices

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- Predictions: consistent with horizon-dependent response of long forwards to short rates
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- Risk-sharing: quantitatively important
  - ⚠ SCF $\times$ term premium estimates: risk-sharing cost $\geq$ output, inflation vol.
  - ⚠ CEX: consumption responds to premium dispersion

Optimal Policy Findings

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  - Calibration shows risk-shaping motive is powerful
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- **Triple mandate**: stable prices, output, and term premia

Related Literature

- Preferred habitat and limits to arbitrage
  - Vayanos & Vila (2021), Greenwood & Vayanos (2014), King (2019), Kekre, Lenel & Mainardi (2024), Ray, Droste & Gorodnichenko (2024); evidence: Krishnamurthy & Vissing-Jorgensen (2011, 2012), D'Amico & King (2013), Di Maggio et al. (2020), ...
- Optimal conventional and unconventional policy; QE in general equilibrium
  - Alexandrov & Brunnermeier (2026), Sims, Wu & Zhang (2023), Silva (2026); Gertler & Karadi (2011, 2013), Curdia & Woodford (2011), Sims & Wu (2020), Karadi & Nakov (2020), ...
- Heterogeneous households and monetary policy
  - Kaplan, Moll & Violante (2018), Auclert (2019), Bhandari et al. (2021), Acharya, Challe & Dogra (2023), Kekre & Lenel (2022), Caballero & Simsek (2024), Campbell & Nemtyrev (2026), ...
- Macro-finance term structure; international
  - Ang & Piazzesi (2003), Piazzesi (2005); Itskhoki & Mukhin (2023), Gourinchas, Ray & Vayanos (2025), Greenwood et al. (2023), Fontanier & Simsek (2026)

Model

Model Overview

- Continuous time New Keynesian model with Vayanos–Vila **bond markets**:
  - Continuum of zero-coupon bonds, maturity $\tau \in [0, T]$, price $P_t^{(\tau)}$
  - Aggregate risk factors $\mathrm{d}z_t = -\kappa_z z_t \,\mathrm{d}t + \sigma_z \,\mathrm{d}B_t$; baseline: technology $Z_t = Ze^{z_t}$
- **Households**: work, consume, save and borrow through a fixed-weight bond fund
  - ⚠ Heterogeneous duration exposure (mortgages vs. deposits; funds rebalance sluggishly)
- **Firms**: monopolistic competitors, Rotemberg price adjustment costs
  - ⚠ Working capital: a portion of the wage bill borrowed in advance, portfolio weights $\beta(\tau)$
- **Arbitrageurs**: trade across the term structure
  - ⚠ Limited risk-bearing capacity: bonds are not priced with household SDF
- **Central bank**: sets the nominal short rate $i_t = \lim_{\tau \to 0} -\log P_t^{(\tau)}/\tau$
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Households

- Household $i \in [0,1]$ maximizes $\mathsf{E}_0 \int_0^\infty e^{-\rho t + \varrho_t} \left(\frac{C_t(i)^{1-\varsigma} - 1}{1-\varsigma} - \frac{N_t(i)^{1+\varphi}}{1+\varphi} \right) \mathrm{d}t$ subject to

$$\mathrm{d}\mathcal{B}_t(i) = [\mathcal{W}_t N_t(i) - \mathcal{P}_t C_t(i)]\,\mathrm{d}t + \mathcal{B}_t(i)\,\mathrm{d}\tilde{\mathcal{R}}_t(i) + \mathrm{d}\mathcal{F}_t$$

- Bond fund return, weights $\eta_i(\tau)$ taken as given ($\int_0^T \eta_i(\tau)\,\mathrm{d}\tau = 1$):

$$\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}} = \mu_t(\tau)\,\mathrm{d}t + \sigma_t(\tau)\,\mathrm{d}B_t\,, \quad \mathrm{d}\tilde{R}_t(i) \equiv \int_0^T \eta_i(\tau) \frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}}\,\mathrm{d}\tau = \tilde{\mu}_t(i)\,\mathrm{d}t + \tilde{\sigma}_t(i)\,\mathrm{d}B_t$$

  - Individually optimal with a fixed cost of rebalancing: portfolio choice is second order

- Intratemporal condition standard; intertemporal condition (log deviations):

$$\mathsf{E}_t\,\mathrm{d}c_t(i) = \varsigma^{-1}\left[\tilde{\mu}_t(i) - \pi_t - \kappa_\varrho \varrho_t + \xi_t(i)\right]\mathrm{d}t$$

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Firms

- Intermediate firms $j \in [0,1]$, $Y_t(j) = Z_t L_t(j)$; CES aggregation with elasticity ϵ_t
- Flow profits and costs of production:

$$\mathrm{d}\Pi_t(j) = \left[(1+\tau^y)\mathcal{P}_t(j)Y_t(j) - \mathcal{W}_t L_t(j) - \mathcal{T}_t^y\right]\mathrm{d}t - \mathrm{d}\Theta_t(j)$$

$$\mathrm{d}\Theta_t(j) = \underbrace{\frac{\vartheta}{2}\pi_t(j)^2\mathcal{P}_t Y_t\,\mathrm{d}t}_{\text{Rotemberg}} + \underbrace{\mathcal{W}_t L_t(j)\,\mathrm{d}\check{R}_t}_{\text{working capital}}, \quad \mathrm{d}\check{R}_t \equiv \int_0^T \beta_t(\tau)\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}}\,\mathrm{d}\tau = \hat{\mu}_t\,\mathrm{d}t + \hat{\sigma}_t\,\mathrm{d}B_t$$

  - $\beta_t = \int_0^T \beta_t(\tau)\,\mathrm{d}\tau$: fraction of the wage bill financed in advance
- Symmetric equilibrium: a familiar Phillips curve with $\mathrm{MC}_t = (1+\hat{\mu}_t)W_t/Z_t$
  - Standard conditional on marginal costs; the departure is the duration-weighted borrowing cost
- ⚠ Firm borrowing: elastic bond demand which makes the short rate move risk prices

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Arbitrageurs, Government, and Asset Market Clearing

- Representative arbitrageur, mean-variance problem over holdings $\mathcal{X}_t(\tau)$:

$$\max \ \mathsf{E}_t \, \mathrm{d}\mathcal{X}_t - \frac{a_t}{2} \mathsf{Var}_t \, \mathrm{d}\mathcal{X}_t \,, \quad \mathrm{d}\mathcal{X}_t = \mathcal{X}_t i_t \, \mathrm{d}t + \int_0^T \mathcal{X}_t(\tau) \left(\frac{\mathrm{d}P_t^{(\tau)}}{P_t^{(\tau)}} - i_t \, \mathrm{d}t \right) \mathrm{d}\tau$$

  - a_t: risk aversion; more generally, capital or VaR constraints, any limit to risk-bearing capacity

- Expected excess returns: return covariances scaled by a_t and holdings

$$\mu_t(\tau) - i_t = a_t \int_0^T \mathcal{X}_t(\tau') \sigma_t(\tau) \sigma_t(\tau')^\top \, \mathrm{d}\tau'$$

- Government: net bond supply $G_t(\tau) = g_t(\tau) - q e_t(\tau)$, fiscal issuance net of QE
  - Policy costs: $\frac{\psi'}{2}(i_t - \bar{i}_t)^2$ (lower bound) and $\int_0^T \frac{\psi(\tau)}{2} \mathrm{QE}_t(\tau)^2 \, \mathrm{d}\tau$ (holding)

- Bond market clearing: $\mathcal{X}_t(\tau) - \beta_t(\tau) \mathcal{W}_t N_t + \int_0^1 \eta_{it}(\tau) \mathcal{B}_t(i) \, \mathrm{d}i = \mathcal{G}_t(\tau)$

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Perturbation and First Best

- Family of economies indexed by ξ: $\sigma_z(\xi) = \xi\sigma_z$, $a(\xi) = \xi^{-2}a$, so $a(\xi)\sigma_z(\xi)\sigma_z(\xi)^\top \to a\sigma_z\sigma_z^\top$
  - "Low risk, low risk-bearing capacity": risk premia survive at first order

Proposition (First-best allocation)

In the limit $\xi \to 0$, with zero net steady-state wealth ($b = 0$) and initially equalized wealth:

- With perfect arbitrage ($a = 0$), the model is equivalent to a representative agent model
- Optimal subsidy, no markup or financing shocks: first-best under flexible prices, $i_t = \rho$

- Steady state: zero inflation, zero arbitrageur positions, consumption equalized
- Arbitrageur optimality: endogenous loadings $\sigma(\tau)$ × global risk prices Λ_t

$$\mu_t(\tau) - i_t = a\sigma(\tau)\Lambda_t \equiv \lambda_t(\tau), \quad \Lambda_t = \int_0^T \sigma(\tau)^\top \chi_t(\tau)\,\mathrm{d}\tau$$

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Aggregate Dynamics and Social Loss

- Modified aggregate Euler equation and Phillips curve (output gap x_t, natural rate v_t):

$$\mathsf{E}_t\,\mathrm{d}x_t = \varsigma^{-1}\left[\tilde{\mu}_t - \pi_t - v_t\right]\mathrm{d}t\,,\quad \tilde{\mu}_t = \int_0^T \eta(\tau)\mu_t(\tau)\,\mathrm{d}\tau\,,\quad \eta(\tau) \equiv \int_0^1 \eta_i(\tau)\,\mathrm{d}i$$

$$\mathsf{E}_t\,\mathrm{d}\pi_t = \left[\rho\pi_t - \kappa\left(x_t + \alpha_{\hat{\mu}}\hat{\mu}_t + \alpha_{\hat{\beta}}\hat{\beta}_t\right) - u_t\right]\mathrm{d}t\,,\quad \hat{\mu}_t = \int_0^T \beta(\tau)\mu_t(\tau)\,\mathrm{d}\tau$$

  - Household and firm effective borrowing rates $\tilde{\mu}_t$, $\hat{\mu}_t$; markup shocks u_t; financing shocks $\hat{\beta}_t$

- Second-order expansion of social welfare relative to the first-best:

$$\mathcal{L}_t \equiv \underbrace{(\varsigma+\varphi)x_t^2 + \vartheta\pi_t^2}_{\text{conventional}} + \underbrace{\frac{\varsigma}{\varphi}(\varphi+\varsigma)\,\mathsf{Var}_i\,\check{c}_t(i)}_{\text{consumption dispersion}} + \underbrace{\int_0^T \psi(\tau)\,qe_t(\tau)^2\,\mathrm{d}\tau + \psi^I\left(i_t - \bar{i}_t\right)^2}_{\text{policy-instrument costs}}$$

- ⚠ Relative to RANK: consumption dispersion $\mathsf{Var}_i\,\check{c}_t(i)$, since risk-sharing is imperfect

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Sufficient Statistics for Wealth Heterogeneity

- Exposures: $\eta_i^\sigma \equiv \int_0^T \eta_i(\tau)\sigma(\tau)\,\mathrm{d}\tau$, $\eta^\sigma \equiv \int_0^1 \eta_i^\sigma\,\mathrm{d}i$, $\beta^\sigma \equiv \int_0^T \beta(\tau)\sigma(\tau)\,\mathrm{d}\tau$
  - Effective rates: $\tilde{\mu}_t(i) = i_t + a\eta_i^\sigma \Lambda_t$, $\tilde{\mu}_t = i_t + a\eta^\sigma \Lambda_t$, $\hat{\mu}_t = \beta i_t + a\beta^\sigma \Lambda_t$
- Relative wealth and consumption ($b = 0$, transfers rebate the aggregate saving flow):

$$\mathrm{d}\check{b}_t(i) = \left[\rho\check{b}_t(i) - (1+\varsigma/\varphi)\,\check{c}_t(i)\right]\mathrm{d}t\,, \quad \mathsf{E}_t\,\mathrm{d}\check{c}_t(i) = \varsigma^{-1}a\,(\eta_i^\sigma - \eta^\sigma)\,\Lambda_t\,\mathrm{d}t$$

Theorem (Sufficient statistics)

Let $b_t^\sigma \equiv \int_0^1 (\eta_i^\sigma)^\top \check{b}_t(i)\,\mathrm{d}i$, $\check{c}_t^\sigma \equiv \int_0^1 (\eta_i^\sigma)^\top \check{c}_t(i)\,\mathrm{d}i$, $\check{\Sigma} \equiv \int_0^1 (\eta_i^\sigma - \eta^\sigma)^\top (\eta_i^\sigma - \eta^\sigma)\,\mathrm{d}i$. If initial relative wealth is spanned by exposures:

- $\check{b}_t(i) = (\eta_i^\sigma - \eta^\sigma)\gamma_t^b$, $\check{c}_t(i) = (\eta_i^\sigma - \eta^\sigma)\gamma_t^c$, with $b_t^\sigma = \check{\Sigma}\gamma_t^b$, $\check{c}_t^\sigma = \check{\Sigma}\gamma_t^c$
- $\Lambda_t = (\beta^\sigma)^\top \left[(1+\varsigma+\varphi)x_t + \frac{1+\varphi}{\varsigma+\varphi} z_{1,t}\right] + g_t^\sigma - qe_t^\sigma + \hat{\beta}_t^\sigma - b_t^\sigma$

- Only risk exposure matters, not specific bonds: the cross-section is $(b_t^\sigma, \check{c}_t^\sigma)$

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  - Effective rates: $\tilde{\mu}_t(i) = i_t + a\eta_i^\sigma \Lambda_t$, $\tilde{\mu}_t = i_t + a\eta^\sigma \Lambda_t$, $\hat{\mu}_t = \beta i_t + a\beta^\sigma \Lambda_t$
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$$\mathrm{d}\check{b}_t(i) = \left[\rho\check{b}_t(i) - (1+\varsigma/\varphi)\,\check{c}_t(i)\right]\mathrm{d}t\,, \quad \mathsf{E}_t\,\mathrm{d}\check{c}_t(i) = \varsigma^{-1}a\,(\eta_i^\sigma - \eta^\sigma)\,\Lambda_t\,\mathrm{d}t$$

Theorem (Sufficient statistics)

Let $b_t^\sigma \equiv \int_0^1 (\eta_i^\sigma)^\top \check{b}_t(i)\,\mathrm{d}i$, $\check{c}_t^\sigma \equiv \int_0^1 (\eta_i^\sigma)^\top \check{c}_t(i)\,\mathrm{d}i$, $\check{\Sigma} \equiv \int_0^1 (\eta_i^\sigma - \eta^\sigma)^\top (\eta_i^\sigma - \eta^\sigma)\,\mathrm{d}i$. If initial relative wealth is spanned by exposures:

- $\check{b}_t(i) = (\eta_i^\sigma - \eta^\sigma)\gamma_t^b$, $\check{c}_t(i) = (\eta_i^\sigma - \eta^\sigma)\gamma_t^c$, with $b_t^\sigma = \check{\Sigma}\gamma_t^b$, $\check{c}_t^\sigma = \check{\Sigma}\gamma_t^c$
- $\Lambda_t = (\beta^\sigma)^\top \left[(1+\varsigma+\varphi)x_t + \frac{1+\varphi}{\varsigma+\varphi} z_{1,t}\right] + g_t^\sigma - qe_t^\sigma + \hat{\beta}_t^\sigma - b_t^\sigma$

- Only risk exposure matters, not specific bonds: the cross-section is $(b_t^\sigma, \check{c}_t^\sigma)$

Dynamics: Analytical Results

Rigid Prices, One Risk Factor

- Natural rate shocks $v_t \equiv -\varsigma\kappa_z \frac{1+\varphi}{\varsigma+\varphi} z_t$ only; fully rigid prices ($\vartheta \to \infty$); Taylor-type rules

$$\mathsf{E}_t\, \mathrm{d}x_t = \varsigma^{-1}\left[i_t + a\eta^\sigma \lambda_t - v_t\right]\mathrm{d}t\,, \quad \mathsf{E}_t\, \mathrm{d}\check{c}_t^\sigma = \varsigma^{-1} a \check{\Sigma} \lambda_t\, \mathrm{d}t$$
$$\mathrm{d}b_t^\sigma = \left[-\left(1+\varsigma/\varphi\right)\check{c}_t^\sigma + \rho b_t^\sigma\right]\mathrm{d}t\,, \quad \lambda_t = -qe_t^\sigma + \beta^\sigma\left((1+\varsigma+\varphi)x_t - v_t/(\varsigma\kappa_z)\right) - b_t^\sigma$$
$$i_t = \phi_x x_t + \epsilon_{i,t}, \quad qe_t^\sigma = \phi_x^\sigma x_t + \epsilon_{q,t}^\sigma$$

Proposition (Rigid price dynamics, no firm borrowing)

Suppose $a > 0$, $\beta(\tau) = 0$, $\phi_x^\sigma = 0$, $\phi_x > 0$, $\eta^\sigma > 0$, and $b_0^\sigma = 0$. Then the model is equivalent to a representative agent model, and $\lambda_t = 0$: $\partial x_t/\partial v_t > 0$, $\partial \lambda_t/\partial v_t = 0$, $\partial x_t/\partial \epsilon_{i,t} < 0$, $\partial \lambda_t/\partial \epsilon_{i,t} = 0$. Following an unconventional policy shock, $\partial x_t/\partial \epsilon_{q,t}^\sigma > 0$, $\partial \lambda_t/\partial \epsilon_{q,t}^\sigma < 0$, and if $\check{\Sigma} > 0$ then $b_{t+k}^\sigma \neq 0$.

- No firm borrowing: zero arbitrageur positions after rate moves, no risk compensation
- QE cuts arbitrageur holdings $\implies$ premia fall, most for long borrowers $\implies$ dispersion

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Rigid Price Dynamics with Firm Borrowing

Proposition (Rigid price dynamics with firm borrowing)

Suppose $\beta^\sigma > 0$, $\eta^\sigma > 0$, $\check{\Sigma} > 0$, $0 < a < \bar{a}$, and $0 < \phi_x < \bar{\phi}_x \equiv \varsigma\kappa_z(\varsigma + \varphi)$.

1. *Technology shock* which decreases the output gap: both i_t and λ_t decline on impact; for all $k > \bar{k}$, $\lambda_{t+k} > 0$ while the short rate remains below steady state; with balance sheet tools ($\phi_x^\sigma < 0$), the impact decline in λ_t is larger
2. *Conventional policy shock* which decreases i_t ($\phi_x^\sigma = 0$): x_t increases; the conditional reaction of λ_t to i_t is negative; smaller magnitude than the unconditional reaction

- Intuition: when $\phi_x < \bar{\phi}_x$, firms reduce borrowing in a recession $\implies$ λ_t declines with i_t
  - Lower premia: long-duration households save less, arbitrageurs go long $\implies$ $\lambda_t > 0$
  - Policy *shock*: an unexpected cut raises firm borrowing $\implies$ conditional correlation negative
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Empirical Evidence

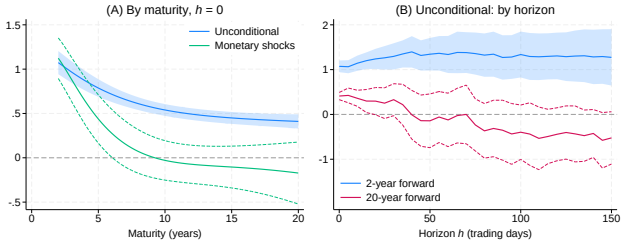
Forward Rates and Short Rates

- Compare with how forward rates respond to short rates (Hanson, Lucca & Wright 2021)
  - Long-dated forwards mostly reflect term premia; no decomposition of yields imposed
- Regressions, τ-year instantaneous forward $f_t^{(\tau)}$, 1-year yield $y_t^{(1)}$, horizon h in trading days:

$$f_{t+h}^{(\tau)} - f_{t-1}^{(\tau)} = \alpha(\tau, h) + \beta(\tau, h)\left(y_t^{(1)} - y_{t-1}^{(1)}\right) + \epsilon_t(\tau, h)$$

  - Daily nominal zero-coupon curves (Gürkaynak, Sack & Wright 2007), 1982 onward
- Unconditional: OLS on all trading days, whatever shocks move short rates
- Monetary shocks: IV on FOMC days, instrumenting with Bauer & Swanson (2023) surprises
- To test: comovement on impact, fading at the long end, shock-dependent, larger with QE

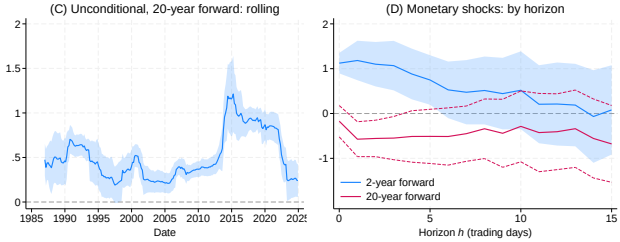
Pass-Through of Short Rates to Forward Rates: Impact and Horizon



$\beta(\tau, h)$: response of the τ-year forward between $t-1$ and $t+h$ to a 1 pp change in the 1-year yield; OLS 1982–2026, IV on FOMC days 1988–2023; bands ± 2 SE.

- Impact: one-for-one at 2 years; the 20-year forward still rises about 40 bp (Proposition 5)
- Short-lived at long maturities: the 20-year response is about zero within three months
- Monetary shocks: pass-through falls faster with maturity, negative beyond 9 years

Pass-Through: State Dependence and Days after FOMC



Panel C: 20-year forward at $h = 0$, rolling 5-year windows dated by window end. Panel D: 2- and 20-year forwards after FOMC announcements, by horizon.

- State dependent: 0.2 to 0.7 before 2011, 1.2 in 2015: stronger with balance sheet tools
- After FOMC days the 20-year response is significantly negative: -0.57 (0.19) at $h = 1$
- Documented by Hanson, Lucca & Wright (2021) and Nakamura & Steinsson (2018) rolling, $h = 60$

Risk-Sharing Quantification

Quantifying the Risk-Sharing Motive

- Consumption dispersion is large: measured $\mathrm{Var}_i\, \check{c}_t(i)$ would dominate the loss
  - Most is unrelated to term premia: we need the part the model's mechanism drives
- Mechanism: premium dispersion on heterogeneous portfolios moves relative consumption

$$\frac{\rho}{2}\,\mathrm{E}_0 \int_0^\infty e^{-\rho t}\mathcal{L}_t\,\mathrm{d}t \approx \frac{\varsigma+\varphi}{2}\,\mathrm{Var}(x_t) + \frac{\vartheta}{2}\,\mathrm{Var}(\pi_t) + \frac{\varsigma}{2\varphi}\,(\varphi+\varsigma)\,\frac{s_0^2\,\mathrm{Var}(\tilde{\lambda}_t)}{\varsigma^2(\rho+\kappa_p)^2}$$

  - $\tilde{\lambda}_t \equiv a\eta^\sigma \Lambda_t$: aggregate household premium
  - s_0: portfolio dispersion (lower bound: differences in duration exposure only, not composition)
- Step 1: measure premium dispersion $s_0 \times \mathrm{SD}(\tilde{\lambda}_t)$ from balance sheets and term premia
- Step 2: map it into consumption dispersion and the loss, against the conventional terms
- Step 3: check the Euler-equation elasticity behind the mapping in household data

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Step 1: Premium Dispersion across Households

- Key object: $s_0\,\mathrm{SD}(\tilde{\lambda}_t) = \mathrm{SD}_i(D_i) \times \mathrm{SD}(\ell_t)$: exposure dispersion $\times$ premium volatility
- $\mathrm{SD}_i(D_i)$: SCF 2022, net duration exposure per dollar of income $=$ 9.0 years
  - Durations: deposits 0, CDs 1, bonds 5, retirement 40/60, mortgages -5, loans -1
- $\mathrm{SD}(\ell_t)$: five-year term premium per year of duration, $\ell_t = \mathrm{TP}_t^{(5)}/2.5$ (ACM; Kim-Wright)
  - κ_p from a monthly AR(1): half-life 3.3 years (ACM), 2.5 years (KW)
- Borrowing-rate dispersion: SD of 303 bp (ACM) or 195 bp (KW)

Step 2: Welfare Losses, Conventional and Risk-Sharing Terms

term	inputs, 1990–2019	loss (% of consumption)
Conventional, $\vartheta = 10$	$\mathsf{SD}(x) = 1.70$, $\mathsf{RMSE}(\pi - 2\%) = 0.67$ (core PCE y/y)	0.05
Conventional, $\vartheta = 50$		0.14
Conventional, $\vartheta = 100$		0.26
Risk-sharing, ACM premia	$\mathsf{SD}(\mathrm{TP}^{(5)}) = 83.8$ bp, $\kappa_p = 0.21$	1.74
Risk-sharing, Kim-Wright premia	$\mathsf{SD}(\mathrm{TP}^{(5)}) = 54.0$ bp, $\kappa_p = 0.28$	0.43

- Conventional terms: $\frac{\varsigma+\varphi}{2}\,\mathsf{Var}(x) + \frac{\vartheta}{2}\,\mathsf{E}[(\pi - 2\%)^2]$, in percent of steady-state consumption
- Risk-sharing term: $\mathsf{SD}_i(\check{c}) = \mathsf{SD}_i(D_i)\,\mathsf{SD}(\ell)/(\varsigma(\rho + \kappa_p))$ gives 13.2% (ACM), 6.5% (KW)
  - 1.74 and 0.43 vs. 0.05 to 0.26: comparable to or larger than the conventional terms
- Conservative: uses risk prices and exposures, not measured consumption dispersion
  - See paper: κ_p-consistent, equity duration, other asset classes all ↑ risk-sharing loss

Step 3: The Euler-Equation Channel in Household Data

- SCF and term premia imply a large motive. Does this reach consumption?
  - Literature: some households are "off Euler equations" (real-rate effects zero or negative)
- Now use the CEX to study our mechanism, and compare with the textbook channel:

$$\Delta \ln c_{i,t} = \alpha_t + \beta D_i \ell_t + \gamma D_i + \delta' X_i + \varepsilon_{i,t}$$

  - Identification: when the premium is high, do exposed households grow consumption faster?
- Data: 1984q1–2012q2, three-quarter windows; 60,599 units, 111 quarters, 28% holders
  - Nondurables and services ex housing and health
  - $D_i = (5 \times \text{savings bonds} + 2.5 \times \text{securities})/\text{income}$, balances dated before the window
- Comparison: real short rate r_t (same for every household)

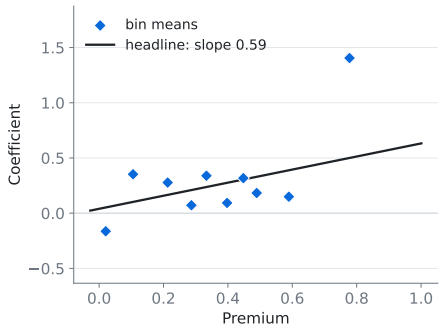
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Household Consumption and the Term-Premium Dispersion



Notes: binned scatter of the quarter-by-quarter coefficient on D_i against the premium, ten bins; line: the headline slope 0.59.

- Recall model: $\beta = \varsigma^{-1}$, baseline $= 1$
- Elasticity on dispersion: **0.59** (0.30)
- Real rate, same households: -0.14 (0.13)
  - Holders vs. non-holders (Vissing-Jorgensen 2002): 0.01 (0.17)
  - Hall (1988); Campbell & Mankiw (1989)
- If anything, the premium channel is stronger
- Robust across consumption measures, sample halves and premium series table
- The risk-sharing-relevant consumption mechanism is at least as large as the textbook Euler channel

Optimal Policy

Simple Policy Rules

- Rigid prices, natural rate shocks; simple rules: functions of the natural states (v_t, b_t^σ)

$$i_t = \chi_{i,v} v_t + \chi_{i,b} b_t^\sigma, \quad qe_t(\tau) = \chi_{q,v}(\tau) v_t + \chi_{q,b}(\tau) b_t^\sigma \implies qe_t^\sigma = \chi_{q,v}^\sigma v_t + \chi_{q,b}^\sigma b_t^\sigma$$

- Policymaker minimizes the unconditional social loss $\mathcal{W} \equiv \frac{1}{2} \mathsf{E} \int_0^\infty e^{-\rho t} \mathcal{L}_t \, \mathrm{d}t$

Lemma (Optimal QE maturity profile)

Dynamics and welfare depend on $\{qe_t(\tau)\}$ only through qe_t^σ and the holding cost; given qe_t^σ, the optimal portfolio is $qe_t(\tau) = \sigma(\tau) \Omega_\psi^{-1} qe_t^\sigma / \psi(\tau)$, with cost $qe_t^{\sigma\top} \Omega_\psi^{-1} qe_t^\sigma$.

- Localization: a continuum of instruments, but only the price of risk is controlled
- Risk-neutral benchmark ($a = 0$): expectations hypothesis, $\mu_t(\tau) = i_t$; RANK
  - $\chi_{i,v} = 1$ achieves the first-best; QE neutrality

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Assume $a > 0$, $\eta^\sigma > 0$, $\beta^\sigma > 0$. If $\check{\Sigma} = 0$, then $\chi^n_{i,v} = 1 + \frac{a\eta^\sigma\beta^\sigma}{\varsigma\kappa_z} > 1$, $\chi_{i,b} = 0$ guarantee $x_t = 0$. With $\check{\Sigma} > 0$, the rule which closes the output gap is not optimal: marginally reducing $\chi_{i,v}$ below $\chi^n_{i,v}$ strictly improves welfare. The optimal rule implies:

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Proposition (Optimal policy separation principle)

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1. Macroeconomic stabilization: $x_t = 0\ \forall t$
2. Term premium stabilization: $\mu_t(\tau) = \tilde{\mu}_t\ \forall \tau$
3. Consumption equalization: $c_t(i) = c_t(i')\ \forall i, i'$

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- QE must sacrifice term premium stabilization to partially stabilize the macroeconomy
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Without Divine Coincidence

- Sticky prices, markup shocks, working capital via $\hat{\mu}_t$; wedge $\Delta^\sigma \equiv \beta^\sigma - \beta\eta^\sigma$

Proposition (Exact macro stabilization)

Suppose $a > 0$, $\beta > 0$, $\beta^\sigma > 0$, $\eta^\sigma > 0$.

1. Whenever $\Delta^\sigma \neq 0$, $x_t \equiv \pi_t \equiv 0$ is an equilibrium under $\Lambda_t = -\tilde{u}_t/(a\Delta^\sigma)$, $i_t = v_t + \frac{\eta^\sigma}{\Delta^\sigma}\tilde{u}_t$, $qe_t^\sigma = -\frac{\beta^\sigma}{\varsigma\kappa_z}v_t - b_t^\sigma - \Lambda_t$, with $\tilde{u}_t = \beta v_t + \frac{\vartheta}{\epsilon}u_t$
2. If $\check{\Sigma} > 0$ and $\tilde{u}_t \not\equiv 0$, as $\Delta^\sigma \to 0$ welfare costs and the policy tools become unbounded

- Two tools target the two effective rates separately and offset any cost-push shock
- But it moves risk prices and worsens risk-sharing; premia explode as $\Delta^\sigma \to 0$

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Proposition (Exact macro stabilization)

Suppose $a > 0$, $\beta > 0$, $\beta^\sigma > 0$, $\eta^\sigma > 0$.

1. Whenever $\Delta^\sigma \neq 0$, $x_t \equiv \pi_t \equiv 0$ is an equilibrium under $\Lambda_t = -\tilde{u}_t/(a\Delta^\sigma)$, $i_t = v_t + \frac{\eta^\sigma}{\Delta^\sigma}\tilde{u}_t$, $qe_t^\sigma = -\frac{\beta^\sigma}{\varsigma\kappa_z}v_t - b_t^\sigma - \Lambda_t$, with $\tilde{u}_t = \beta v_t + \frac{\vartheta}{\epsilon}u_t$
2. If $\check{\Sigma} > 0$ and $\tilde{u}_t \not\equiv 0$, as $\Delta^\sigma \to 0$ welfare costs and the policy tools become unbounded

- Two tools target the two effective rates separately and offset any cost-push shock
- But it moves risk prices and worsens risk-sharing; premia explode as $\Delta^\sigma \to 0$

Full Commitment

- Both instruments depend on the entire history of shocks; the planner minimizes $\mathcal{W}$
  - Class: time-invariant feedback on innovations, via policy memory states m_t
  - $i_t = \phi_i^\top s_t + \phi_{im}^\top m_t + \epsilon_{i,t}$, $qe_t^\sigma = \Phi_q s_t + \Phi_{qm} m_t + \epsilon_{q,t}$; the paths ϵ move returns, not loadings
- First result: at fixed composites, a standard LQ program with promise-keeping costates θ_t
- Nash plan: the textbook plan at the composites it induces; exposures taken as given
- Stackelberg plan: a leader in the bond market, internalizing how it shapes $\sigma(\tau)$
- Shared benefit of commitment: inertial, price-level-targeting-style responses
  - Here, smaller *ex ante* variation in policy tools also lowers bond volatility and risk compensation

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Full Commitment: Risk-Shaping

Theorem (Full commitment optimal policy, necessary conditions)

Assume $\psi^i > 0$. Let $\nu(\tau)$ be the multipliers on the endogenous loadings $\sigma(\tau)$ and define shadow factors $\upsilon_t^i \equiv \int_0^\infty \nu_i(u)^\top \, \mathrm{d}B_{t-u}$, $\upsilon_t^\Lambda \equiv \int_0^\infty \nu_\Lambda(u)^\top \, \mathrm{d}B_{t-u}$ (ν_i, ν_Λ: ν integrated over the bonds outstanding at horizon u). Any optimal plan satisfies the Nash system with $\theta_{\Lambda,t} - \upsilon_t^\Lambda$ in place of $\theta_{\Lambda,t}$ and $-\upsilon_t^i$ added to the short-rate condition. The multiplier function is spanned by K basis functions with coefficients $\zeta_k = \partial J / \partial c_k$; with a finite-state realization, the plan solves a finite-dimensional fixed point in (c, ζ, M). Instruments are $\psi^i(i_t - \bar{i}_t) = \kappa\alpha_{\hat{\mu}}\beta\theta_{\pi,t} - \varsigma^{-1}\theta_{x,t} + \upsilon_t^i$, $qe_t^\sigma = \Omega_\psi(\theta_{\Lambda,t} - \upsilon_t^\Lambda)$; Nash sets $\upsilon = 0$.

- Double duty of $\sigma(\tau)$: the risk-adjusted path of policy *is* each bond's loading
  - Sets premia, $\lambda_t(\tau) = a\sigma(\tau)\Lambda_t$, and exposures, $\eta_i^\sigma = \int \eta_i(\tau)\sigma(\tau)\,\mathrm{d}\tau$
- Risk-shaping: rate response to a shock can set a portfolio's exposure/premium to zero
  - Not for all portfolios at once; $\sigma(\tau) \equiv 0$ requires a full rate peg
  - Note: risk-shaping requires knowledge of welfare-relevant portfolios

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Quantitative Analysis

Calibration

- Sticky prices, working capital, three shocks: demand, markup and bond supply
- Standard NK values, $\vartheta = 100$; Taylor rule $i_t = 1.5\pi_t + 0.5x_t$
- Firms finance 3/4 of the wage bill at a mean maturity of 5 years
- Households: half hold $1 - s_0$, half $1 + s_0$ of a 5-year portfolio; $s_0 D_h = 9$ years (SCF)
- Internal: shock volatilities, demand-supply correlation, Phillips curve slope, a
  - Targets: SD and corr of x, π; premium SD 84 bp (ACM); $\mathrm{corr}(\Delta_{12} i, \Delta_{12}\ \text{prem})$
- Under the Taylor rule, consumption dispersion is 60% of the loss (0.188 of 0.293) tables

Social Loss under Simple Rules

	loss (% of consumption)						
	total	x	π	dispersion	rate	prem.	u-resp.
Historical Taylor rule	0.309	0.034	0.071	0.188	0.016	84	1.52
Taylor, $\phi_\pi = 3$	0.361	0.028	0.062	0.248	0.022	100	2.54
Gap-stabilizing Taylor	0.357	0.000	0.046	0.283	0.028	109	1.17
Inflation-stabilizing Taylor	0.469	0.028	0.055	0.353	0.033	126	3.82
Natural-rate rule, $\chi_{i,v} = 1$	0.406	0.021	0.051	0.310	0.024	116	1.52
Optimal simple rule	0.266	0.033	0.079	0.141	0.014	70	0.22
Historical rule, $\Lambda = 0$	0.098	0.021	0.066	0.000	0.011	0	1.52
Natural-rate rule, $\Lambda = 0$	0.061	0.000	0.045	0.000	0.016	0	1.52

Loss $\frac{1}{2}\,\mathbf{E}[\mathcal{L}_t]$, % of consumption, rate cost $\psi^i = 1.87$ throughout; $\Lambda = 0$: costless balance sheet holding the price of risk at zero; prem.: SD of the household premium (bp).

- Aggressive rules are worse: lower macro but higher premium volatility and dispersion
- Optimal simple rule improves modestly, mainly by responding much less to markup shocks
- Costless QE holding $\Lambda_t = 0$ removes dispersion; what remains is mostly inflation

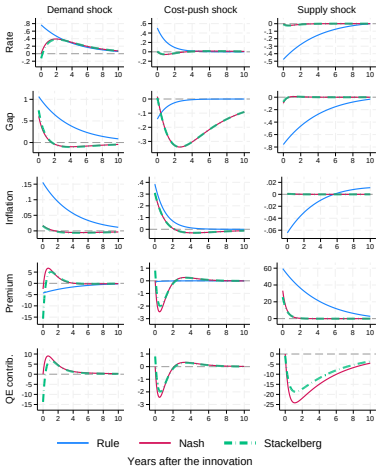
Social Loss under Commitment

	loss (% of consumption)							
	total	x	π	dispersion	rate	QE	prem.	u-resp.
Historical Taylor rule	0.309	0.034	0.071	0.188	0.016	–	84	1.52
Nash plan, no QE	0.110	0.008	0.023	0.072	0.008	–	46	−0.23
Stackelberg plan, no QE	0.047	0.015	0.023	0.002	0.007	–	6	−0.10
Nash plan, low QE cost	0.038	0.007	0.023	<0.001	0.006	0.002	17	−0.13
Stackelberg plan, low QE cost	0.038	0.008	0.023	<0.001	0.006	0.001	15	−0.11

Same units and rate cost; the low QE cost is 1/240 of the cost at which QE halves dispersion under the Nash plan.

- Commitment is valuable: the Nash plan cuts the loss by two thirds, mostly dispersion
- Stackelberg more than halves the rest, all of it dispersion: it chooses exposures
- Rate-only Stackelberg beats the $\Lambda = 0$ rules: exposure shaping substitutes for $\Lambda = 0$
- With cheap QE the two plans coincide: little left for risk-shaping to do

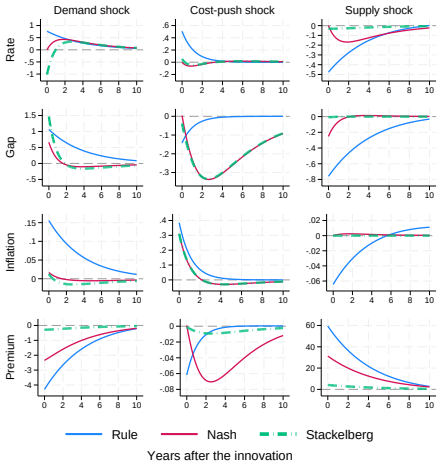
Impulse Responses under Optimal Policy: Short Rate and Balance Sheet



One-SD shocks; low balance sheet cost. Rate in pp, gap and inflation in percent, household premium and the balance sheet's contribution to it in bp.

- Supply shock: separation principle; purchases build over the first year and eliminate the premium
- Short rate follows the commitment path: gradual, stabilizes through expected future policy
- Demand shock: balance sheet raises the premium at intermediate horizons: partly a macro stabilization tool; the plans differ mainly in the first months

Risk-Shaping through the Path of the Short Rate



Short rate only, $\psi^i = 1.87$; same units.

- Demand shock: Stackelberg plan combines a short-lived cut with the gradual tightening of the Nash plan
- Long bond returns depend on the whole expected path: short bonds gain, long bonds lose; the household portfolio is much less exposed
- Costly rate: the Nash plan becomes much less reactive and the gap between the plans closes; cheap QE then takes on macro stabilization

IRFs

Conclusion

Concluding Remarks

- GE model with market segmentation, nominal frictions, and household heterogeneity
  - Household distribution summarized by a small number of risk-weighted sufficient statistics
- Losses from inflation, output, and consumption dispersion induced by term premia
  - SCF: comparable to or larger than conventional terms; CEX: the channel is present
- **Separation principle**: short rate to output and inflation, balance sheet to term premia
  - Constraints on either instrument imply trade-offs between the two objectives
- Under commitment, policy also shapes household exposure to aggregate risk
  - A quarter of the commitment gains without QE; simple QE rules get most of them
- **Triple mandate**: stable prices, output, and term premia

Thank You!

Robustness: Duration Supply Responding to the Short Rate

- Firm borrowing is not the only duration supply that moves with the short rate
  - Corporate issuance, mortgage convexity hedging, reaching for yield
- Add a reduced-form duration supply loading on the short rate and the state
- A short-rate loading shifts the conditional and unconditional impact responses equally
  - The difference is the output gap's response to the short rate: with the rule, against a shock
- Robust prediction: the ordering, conditional response below the unconditional one
  - Its sign is a quantitative question; Section 4 tests the ordering (0.41 vs. −0.17)

back

Duration Supply Responding to the Short Rate: Result

- Other duration supply moves with the short rate: issuance, hedging, yield-seeking
- Reduced form in the risk price: $\lambda_t = \beta^\sigma(\cdots) + d_t - b_t^\sigma$, $d_t = \theta_i i_t + \theta_x x_t + \theta_z z_t$
  - $\theta_i > 0$: a lower short rate reduces the duration arbitrageurs absorb, *whatever moved it*

Proposition (Impact responses with reduced-form duration supply)

At $a = 0$, per unit of the impact short-rate change, the two impact responses satisfy

$$\Lambda^u - \Lambda^c = \tilde{\beta}\left[\frac{1}{\phi_x} + \frac{1}{\varsigma\kappa_i}\right] - \frac{\tilde{\theta}_z}{\phi_x|\Omega_{xz}|}, \quad \tilde{\beta} \equiv \beta^\sigma(1 + \varsigma + \varphi) + \theta_x,$$

independent of θ_i and positive under the conditions of Proposition 5; $\Lambda^c < 0$ iff $\theta_i < \tilde{\beta}/(\varsigma\kappa_i)$.

- A direct loading shifts both responses equally: the robust prediction is the ordering

back

CEX: Perturbations and the Mapping to the Risk-Sharing Term

	$\hat{\beta}$ (SE)	holders vs. non-holders
headline	0.59 (0.30)	0.01 (0.17)
strict nondurables	0.53 (0.34)	−0.13 (0.24)
1984–1998	0.95 (0.62)	0.26 (0.42)
1999–2012	0.33 (0.32)	−0.32 (0.27)
Kim–Wright premium	1.26 (0.58)	−0.06 (0.21)

Notes: headline: nondurables and services excluding health, unweighted, interview-month premium. Kim–Wright: pairs from 1990q1 (ACM on the same pairs: 0.48). SEs clustered by quarter.

- Textbook column: the same regression with $r_t \times$ holder status (Vissing-Jorgensen 2002)
  - Do holders raise consumption growth with r_t more than non-holders? Prediction: $1/\varsigma$
- Finding: our test supports the Euler-equation mechanism; the textbook test finds nothing
  - Across consumption measures, sample halves and premium series
- The channel the risk-sharing term relies on is the one that shows up in household data

back

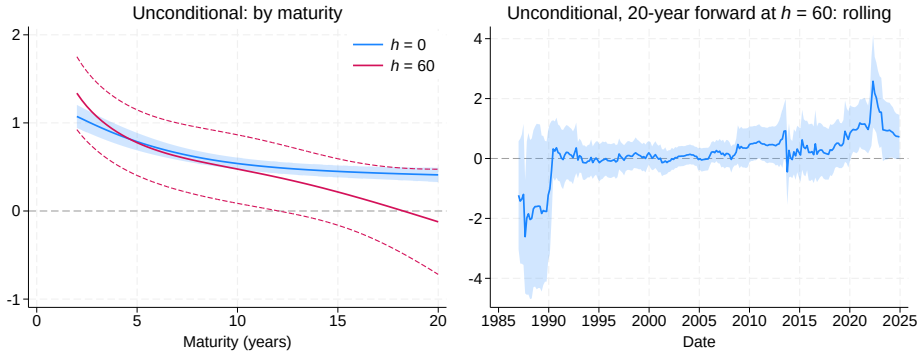
CEX: The Mapping to the Risk-Sharing Term

	$\hat{\beta}$ (SE)	$\varsigma = 1/\hat{\beta}$	$\mathrm{SD}_i(\hat{c})$, %		dispersion term, %		conventional
			ACM	KW	ACM	KW	$\vartheta = 10/50/100$
paper, $\varsigma = 1$	1.00	1.00	13.1	6.5	1.74	0.43	0.06 / 0.18 / 0.33
headline (end of the interview month)	0.59 (0.30)	1.69	7.8	3.8	1.38	0.34	0.07 / 0.19 / 0.34
end of quarter q	0.79 (0.30)	1.27	10.3	5.1	1.55	0.38	0.06 / 0.18 / 0.33
half 1984–1998	0.95 (0.62)	1.05	12.5	6.2	1.70	0.42	0.06 / 0.18 / 0.33
half 1999–2012q2	0.33 (0.32)	3.04	4.3	2.1	1.16	0.29	0.09 / 0.21 / 0.36
headline, directly held floor (12% of $\mathrm{Var}_i(D)$)	0.59 (0.30)	1.69	2.7	1.3	0.17	0.04	0.07 / 0.19 / 0.34
headline, directly held floor (23% of $\mathrm{Var}_i(D)$)	0.59 (0.30)	1.69	3.7	1.8	0.32	0.08	0.07 / 0.19 / 0.34

$\mathrm{SD}_i(\hat{c}) = \hat{\beta} \times \mathrm{SD}_i(D_i) \times \mathrm{SD}(\ell)/(\rho + \kappa_p)$ with $\mathrm{SD}_i(D_i) = 9.0$, $\mathrm{SD}(\ell) = 0.335$ pp (ACM) or 0.216 (KW), $\kappa_p = 0.21$ (ACM) or 0.28 (KW). Dispersion term $\varsigma(\varsigma + \varphi)/(2\varphi) \times \mathrm{SD}_i(\hat{c})^2$ at $\varsigma = 1/\hat{\beta}$, $\varphi = 1$; conventional terms at the same ς. The directly held floor replaces $\mathrm{SD}_i(D_i)$ by its directly held component (bonds, CDs and funds).

back

Forward Rates: Full Maturity Profile and Rolling 60-Day Response



Left: unconditional $\beta(\tau, h)$ by maturity at $h = 0$ and $h = 60$. Right: 20-year forward at $h = 60$, rolling 5-year windows: negative before 1991, near zero 1991–2008, positive after 2020; the sign reversal depends on the sample.

back

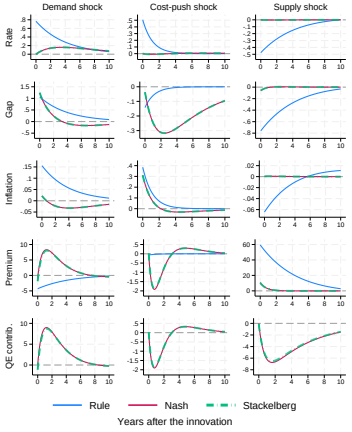
Calibration: Parameters and Moments

Panel A: External		
ρ	discount rate	0.02
ς	inverse elasticity of intertemporal substitution	1
φ	inverse Frisch elasticity	1
ϑ	price-adjustment cost (loss weight on inflation)	100
β	share of the wage bill financed by borrowing	0.75
D_f	maturity of firm borrowing (years)	5
$D_h = 1/\delta_h$	maturity of the household bond portfolio (years)	5
s_0	dispersion of household duration exposure	1.8
κ_g	mean reversion, demand factor	0.25
κ_u	mean reversion, cost-push factor	0.85
κ_s	mean reversion, bond-supply factor	0.20
$1/\delta_s$	maturity of bond supply (years)	5.4
ζ_s	scale of bond supply	3
ϕ_π	historical rule: response to inflation	1.5
ϕ_x	historical rule: response to the output gap	0.5
Panel B: Internal		
σ_g	volatility, demand innovation	0.0333
σ_u	volatility, cost-push innovation	0.00334
ρ_{gs}	correlation, demand and supply innovations	-0.079
κ	slope of the Phillips curve	0.0319
a	arbitrageur risk aversion	6.63

	Data	Model
Panel A: Targeted moments		
SD(x) (%)	1.85	1.85
SD(π), core PCE y/y (%)	0.38	0.38
corr(x, π)	0.53	0.53
SD(household term premium) (bp)	84	84
corr($\Delta_{12} i$, Δ_{12} premium)	-0.53	-0.53
Panel B: Untargeted moments		
SD(i) (%)	–	1.31
SD(2-year term premium) (bp)	–	56
SD(10-year term premium) (bp)	–	144
Panel C: Loss under the historical rule (% of consumption)		
output gap	–	0.034
inflation	–	0.071
consumption dispersion	–	0.188
total	–	0.293

back

Costly Short Rate, Cheap Balance Sheet



- After a demand shock, the balance sheet raises the household premium for several years to tighten financial conditions
- Premium volatility higher than without balance sheet policy

back